Notes

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1 Estimating continuous-valued neural representational geometry with denoising score matching

1.1 Introduction

Representational similarity analysis is widely used in neuroscience and cognitive science to characterize how neural populations encode stimuli by estimating a representational dissimilarity matrix. Existing approaches have provided useful tools for this purpose, including correlation-based distances, Euclidean distances. However, these methods does not account for the noise structure of the data, and is variant to subjective choices of the coordinate system. Mahalanobis distances and decoding-based measures exploit the statistical nature of the problem. However, in order to estimate covariance or decoding, data must be binned first. Binning breaks the continuous nature of the problem, furthermore, when the number of data is sparse, binning reduce the number of data points even more, reducing the method reliability. Other methods like Wishart process does not binning the data yet requires explicitly assume the statistical model of the data (e.g. simple Gaussian) which is often to simplistic.

Here we propose a score-based continuous-stimulus method for estimating representational geometry directly from neural responses without binning the stimulus variable. Our method exploits the smooth structure of the stimulus axis by estimating the conditional score of the neural response distribution with respect to stimulus using score matching (a type of diffusion model), and then using this quantity to construct pairwise dissimilarities with a direct probabilistic interpretation. This provides a principled framework for recovering representational geometry of continuous-valued stimuli from sparse and noisy neural data.

1.2 Methods

Denoting the continuous-valued stimulus variable as θ, and the neural population response is modeled as a conditional distribution $p(x \mid \theta)$ which is unknown. A dataset is sampled from the joint distribution $p(x, \theta) = p(\theta)p(x \mid \theta)$ where $p(\theta)$ is the prior distribution which is often known by the experimenter (e.g. sampling uniform orientation of the grating image) or easy to estimate because of the low-dimensional nature of the stimulus.

Representation dissimilarity analysis cares about the "distance" between two stimulus conditions, mathematically, the distance between two conditional distributions. A straightforward choice is KL divergence, however, KL divergence is not a real distance metric. Therefore in this project, we use Hellinger distance $H(p_i, p_j)$

$$H^2(p_i, p_j) = 1 - \int \sqrt{p(x \mid \theta_i)\, p(x \mid \theta_j)}\, dx, \qquad p_i := p(x \mid \theta_i), \;\; p_j := p(x \mid \theta_j).$$

Besides being a true distance metric, Hellinger distance is attractive here for two additional reasons. First, it is locally connected to Fisher information. When two stimulus conditions are infinitesimally close, the squared Hellinger distance reduces, to leading order, to the Fisher-information metric. Second, Hellinger distance bounds the performance of the optimal binary classifier. In particular, a larger Hellinger distance implies a smaller achievable Bayes error. For these reasons, Hellinger distance provides a natural probabilistic notion of representational dissimilarity.

To connect Hellinger distance to a quantity that can be estimated from samples, define the log-likelihood ratio

$$\Delta\ell_{ij}(x) := \log p(x \mid \theta_j) - \log p(x \mid \theta_i).$$

Then the Bhattacharyya affinity can be written as

$$\sqrt{p(x \mid \theta_i)\, p(x \mid \theta_j)} = p(x \mid \theta_i) \exp\left(\frac{\Delta\ell_{ij}(x)}{2}\right),$$

which yields

$$H^2(p_i, p_j) = 1 - \mathbb{E}_{x\sim p_i}\left[\exp\left(\frac{\Delta\ell_{ij}(x)}{2}\right)\right].$$

Although exact, this one-sided form is asymmetric at the sample level because the expectation is taken only under p_i. To obtain a symmetric sample-wise quantity, we instead use the identity

$$H^2(p_i, p_j) = \frac{1}{2}\,\mathbb{E}_{x_i\sim p_i,\, x_j\sim p_j}\left[h_{ij}(x_i) + h_{ij}(x_j)\right],$$

where

$$h_{ij}(x) := 1 - \operatorname{sech}\left(\frac{\Delta\ell_{ij}(x)}{2}\right).$$

This form is explicitly symmetric in i and j, and expresses the Hellinger distance as the average of two sample-wise contributions, one evaluated on $x_i \sim p_i$ and the other on $x_j \sim p_j$.

The remaining challenge is therefore to estimate $\Delta\ell_{ij}(x)$ without discretizing θ. To exploit the continuous nature of the problem, we decompose the log-likelihood ratio into integration

$$\Delta\ell_{ij}(x) = \int_{\theta_i}^{\theta_j} \partial_\theta \log p(x \mid \theta)\, d\theta.$$

If we are able to estimate the Fisher score of all data points θ_i, integration can be computed numerically into summation.

How to estimate Fisher score? We use Bayes' rule:

$$\partial_\theta \log p(x \mid \theta) = \partial_\theta \log p(\theta \mid x) - \partial_\theta \log p(\theta).$$

The gradient of prior distribution can be computed analytically if the prior distribution is known by the experimentalist; if not known, estimating is relatively straightforward due to the low-dimensional nature of the stimulus.

We use denoising score matching to estimate $\partial_\theta \log p(\theta \mid x)$. The detail is written in previous section.

Overall, the pipeline is the following: Given a dataset $\{(\theta_i, x_i)\}_{i=1}^N$, we estimate the Fisher score of all data points θ_i using denoising score matching, and then perform path integration to obtain log-likelihood ratio, finally compute $h_{ij}(x)$. Notably, there's no data binning in this pipline. The final H_{ij} is the expecation of $h_{ij}(x)$ over all data points x_i and x_j, which can be estimated by simple binning, or more consistently using Gaussian process regression to estimate without any binning. But for simplicity, we use binning in preliminary results. The details of the pipeline is given in Supplementary Information.

1.3 Preliminary results

We generated a toy 2D linear model where neural tuning is $(x, 2x)$ with independent Gaussian noise std linearly ramped from 0.2 to 4. As an approximation of the ground truth, we generate large dataset with 5000 sample points and then estimated Hellinger distance matrix and pairwise decoding matrix, respectively. Hellinger distance matrix is estimated as stated in the method section; pairwise decoding matrix is estimated by binning the data and use logistic regression.

To study how well each of these two methods performs when the dataset size is small, we subsampled smaller dataset with $n = 50$, sample points. For each of the subsampled dataset, we estimated RDM respectively and then compared the correlation of the off-diagonal entries to the approximated ground truth RDM.

Results showing that our Hellinger distance matrix estimated by denoising score matching method exploit the continuous nature of the problem and converges faster than pairwise decoding method, see Figure 1.

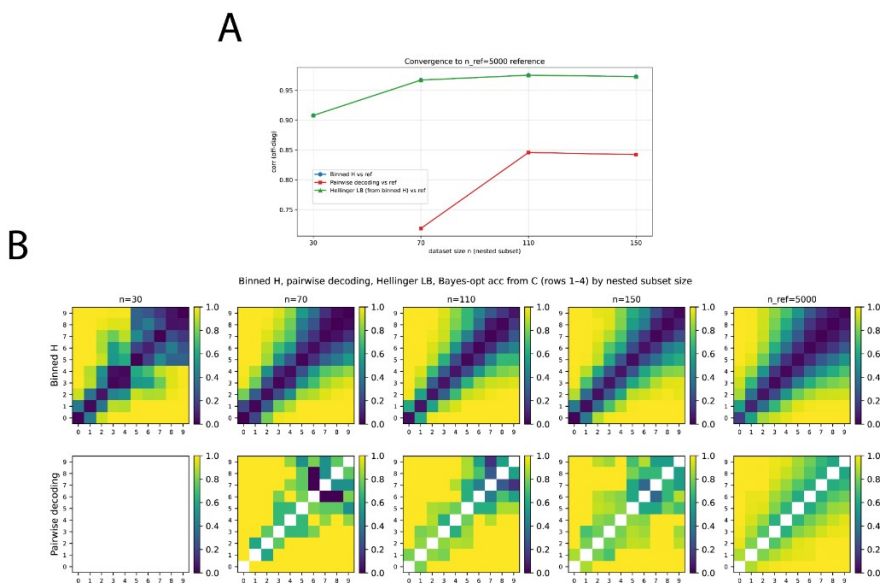

Figure 1: Estimation from subset of dataset convergence of geometry estimates to a large reference ($n_{\text{ref}} = 5000$). **(A)** Correlation of off-diagonal matrix entries to the reference for binned Hellinger (H), pairwise decoding, and a Hellinger lower bound derived from binned H (blue and green overlap), as nested subset size n increases. **(B)** Heatmaps of binned H, pairwise decoding, Hellinger lower bound, and Bayes-optimal accuracy (rows) across $n \in \{30, 70, 110, 150\}$ and the reference column. Pairwise decoding was computed by binning the data and using logistic regression to estimate decoding accuracy. Dataset: linear tuning $(x, 2x)$ with Gaussian noise std linearly ramped from 0.2 to 4.

1.4 Supplementary information: From dataset $\{(\theta_i, x_i)\}_{i=1}^N$ to the h-matrix

We start from a dataset

$$\mathcal{D} = \{(\theta_i, x_i)\}_{i=1}^N,$$

where $\theta_i \in \mathbb{R}$ is a scalar condition and x_i is the corresponding observation. Our goal is to construct an $N \times N$ matrix

$$\mathbf{H}^2 = [H_{ij}^2],$$

whose entry H_{ij}^2 is the sample-wise symmetric contribution of x_i when comparing the conditional distributions $p(x \mid \theta_i)$ and $p(x \mid \theta_j)$.

Step 1: learn posterior and prior scores in θ. Using denoising score matching, we train two score models:

$$s_\phi(\tilde{\theta}, x, \sigma) \approx \partial_\theta \log p_\sigma(\tilde{\theta} \mid x), \qquad r_\psi(\tilde{\theta}, \sigma) \approx \partial_\theta \log p_\sigma(\tilde{\theta}),$$

where p_σ denotes the distribution smoothed in θ by Gaussian noise of scale σ. Concretely, the posterior score model is trained by minimizing

$$\mathcal{L}_{\text{post}}(\phi) = \mathbb{E}_{(\theta,x)\sim\mathcal{D},\ \sigma,\ z\sim\mathcal{N}(0,1)}\Big[\big\|\sigma s_\phi(\theta+\sigma z, x, \sigma) + z\big\|^2\Big],$$

and the prior score model is trained by minimizing

$$\mathcal{L}_{\text{prior}}(\psi) = \mathbb{E}_{\theta\sim\{\theta_i\},\ \sigma,\ z\sim\mathcal{N}(0,1)}\Big[\big\|\sigma r_\psi(\theta+\sigma z, \sigma) + z\big\|^2\Big].$$

For a small evaluation noise level σ_{eval}, define the recovered Fisher score

$$\hat{g}(\theta, x) := s_\phi(\theta, x, \sigma_{\text{eval}}) - r_\psi(\theta, \sigma_{\text{eval}}).$$

By Bayes' rule,

$$\partial_\theta \log p(x \mid \theta) = \partial_\theta \log p(\theta \mid x) - \partial_\theta \log p(\theta),$$

so $\hat{g}(\theta, x)$ approximates the Fisher score $\partial_\theta \log p(x \mid \theta)$.

Step 2: sort the dataset by θ. For notational simplicity, we assume the samples are reordered such that

$$\theta_1 \le \theta_2 \le \cdots \le \theta_N.$$

If the original dataset is not sorted, we sort it by θ, perform all computations below in the sorted order, and permute the final matrices back to the original order if desired.

Step 3: construct the recovered Fisher score matrix. For each sample x_i and each observed condition θ_k, define

$$G_{ik} := \hat{g}(\theta_k, x_i).$$

Thus $\mathbf{G} = [G_{ik}] \in \mathbb{R}^{N\times N}$ is the matrix whose i-th row corresponds to a fixed observation x_i, and whose k-th column evaluates the recovered Fisher score at condition θ_k.

Step 4: construct the cumulative integral matrix. For each row i, define the cumulative trapezoidal integral

$$C_{i1} = 0, \qquad C_{i,k+1} = C_{ik} + \frac{\theta_{k+1} - \theta_k}{2}\big(G_{ik} + G_{i,k+1}\big), \quad k = 1, \ldots, N-1.$$

Equivalently,

$$C_{ij} = \sum_{k=1}^{j-1} \frac{\theta_{k+1} - \theta_k}{2}\big(G_{ik} + G_{i,k+1}\big), \qquad j = 1, \ldots, N.$$

For fixed i, C_{ij} approximates the integral

$$C_{ij} \approx \int_{\theta_1}^{\theta_j} \hat{g}(\theta, x_i)\, d\theta.$$

Step 5: construct the log-likelihood-ratio matrix $\Delta\mathbf{L}$. For a sample (θ_i, x_i), define

$$\Delta\ell_{ij}(x_i) := \log p(x_i \mid \theta_j) - \log p(x_i \mid \theta_i).$$

Since

$$\Delta\ell_{ij}(x_i) = \int_{\theta_i}^{\theta_j} \partial_\theta \log p(x_i \mid \theta)\, d\theta,$$

we estimate it by

$$\Delta L_{ij} := C_{ij} - C_{ii}.$$

Thus the matrix $\Delta\mathbf{L} = [\Delta L_{ij}] \in \mathbb{R}^{N\times N}$ is the estimated log-likelihood-ratio matrix, where

$$\Delta L_{ij} \approx \Delta\ell_{ij}(x_i).$$

By construction,

$$\Delta L_{ii} = 0.$$

Step 6: construct the directed sample-wise contribution matrix. Define

$$H_{ij}^{\rightarrow} := 1 - \operatorname{sech}\left(\frac{\Delta L_{ij}}{2}\right).$$

This is a directed matrix, because its (i, j)-entry is evaluated on x_i only.

Step 7: symmetrize the matrix. To obtain a symmetric pairwise matrix, average the two directions:

$$H_{ij}^{sym} := \frac{1}{2}\left(H_{ij}^{\rightarrow} + H_{ji}^{\rightarrow}\right).$$

Thus the full pipeline is

$$\{(\theta_i, x_i)\}_{i=1}^N \longrightarrow s_\phi,\, r_\psi \longrightarrow \mathbf{G} \longrightarrow \mathbf{C} \longrightarrow \Delta\mathbf{L} \longrightarrow \mathbf{H}^{\rightarrow} \longrightarrow \mathbf{H}^{sym}.$$

Algorithmic summary.

1. Train $s_\phi(\theta, x, \sigma)$ and $r_\psi(\theta, \sigma)$.
2. Compute
$$G_{ik} = s_\phi(\theta_k, x_i, \sigma_{\text{eval}}) - r_\psi(\theta_k, \sigma_{\text{eval}})$$
for all i, k.
3. Compute the cumulative integral matrix $\mathbf{C}$ row by row:
$$C_{i1} = 0, \qquad C_{i,k+1} = C_{ik} + \frac{\theta_{k+1} - \theta_k}{2}(G_{ik} + G_{i,k+1}).$$
4. Compute the directed log-ratio matrix
$$\Delta L_{ij} = C_{ij} - C_{ii}.$$
5. Compute the directed contribution matrix
$$H_{ij}^{\rightarrow} = 1 - \operatorname{sech}\left(\frac{\Delta L_{ij}}{2}\right).$$

6. Symmetrize:

$$H_{ij}^{sym} = \frac{1}{2}\left(H_{ij}^{\rightarrow} + H_{ji}^{\rightarrow}\right).$$

Finally, the Hellinger distance matrix is given by the expectation which can be computed by binning H_{ij}^{sym} or regression method:

$$\mathbf{H}^2 = [\mathbb{E}_{x_i \sim p_i,\, x_j \sim p_j}[H_{ij}^{sym}]].$$

1.5 Supplementary information: Hellinger distance and bounds on classification accuracy

For binary classification between conditions i and j with equal class priors, let

$$p_i(x) = p(x \mid \theta_i), \qquad p_j(x) = p(x \mid \theta_j),$$

and let H_{ij} denote the Hellinger distance between p_i and p_j. The Bayes-optimal classification accuracy is

$$A_{ij}^* = \frac{1}{2}\big(1 + \mathrm{TV}(p_i, p_j)\big),$$

where $\mathrm{TV}(p_i, p_j)$ is the total variation distance. Using the inequalities

$$H_{ij}^2 \le \mathrm{TV}(p_i, p_j) \le \sqrt{2H_{ij}^2 - H_{ij}^4},$$

we obtain

$$\frac{1}{2}(1 + H_{ij}^2) \le A_{ij}^* \le \frac{1}{2}\left(1 + \sqrt{2H_{ij}^2 - H_{ij}^4}\right).$$

Therefore, a measured Hellinger distance directly constrains the best achievable pairwise classification accuracy. In particular, larger Hellinger distance implies better possible discriminability, with $H_{ij} = 0$ corresponding to chance-level accuracy $A_{ij}^* = 1/2$, and $H_{ij} = 1$ corresponding to perfect classification $A_{ij}^* = 1$.

2 Hellinger distance estimation using iterative MDS embedding and flow matching (HeIM-Flow)

2.1 Results

Consider a conditional distribution $p(x \mid \theta)$ with continuous parameter $\theta \in \mathbb{R}^m$. Given paired data (θ, x), we wish to estimate the Hellinger distance between two conditionals, $p(x \mid \theta_1)$ and $p(x \mid \theta_2)$.

A straightforward approach is to bin the data near θ_1 and θ_2, train a binary classifier on x, and convert its output into an estimate of the Hellinger distance. Binning, however, discards the continuity of θ: only samples falling in the chosen bins contribute, and estimates need not vary smoothly as θ_1 and θ_2 move. Kernel weighting avoids hard bins but introduces a fixed bandwidth and still does not provide flexible exploitation of the information from nearby θ.

We propose a method that more directly exploits continuity in θ. Write $p_1 = p(x \mid \theta_1)$, $p_2 = p(x \mid \theta_2)$, and

$$\ell_{12}(x) = \log \frac{p_1(x)}{p_2(x)}. \tag{1}$$

The squared Hellinger distance admits the symmetric representation

$$H^2(p_1, p_2) = \frac{1}{2}\,\mathbb{E}_{X_1 \sim p_1}\left[\psi\big(\ell_{12}(X_1)\big)\right] + \frac{1}{2}\,\mathbb{E}_{X_2 \sim p_2}\left[\psi\big(\ell_{12}(X_2)\big)\right], \qquad \psi(u) := 1 - \operatorname{sech}\Big(\frac{u}{2}\Big). \tag{2}$$

Since sech is an even function, the same function $\psi(\ell_{12}(x))$ can be evaluated for samples from either distribution. Thus, once the log-likelihood ratio $\ell_{12}(x)$ is available, the Hellinger distance can be approximated by Monte Carlo.

Flow matching provides an ODE whose solution yields $\log p(x \mid \theta)$ via the instantaneous change-of-variables formula. A natural baseline is therefore to train a conditional flow on (θ, x): the velocity network takes θ as an explicit conditioning variable, so the model can interpolate across nearby parameter values. Concretely, one fits the flow, evaluates $\log p(x \mid \theta)$, and plugs the resulting log-likelihood ratios into (2) for estimating the Hellinger distance.

This baseline works well when $p(x \mid \theta)$ varies smoothly in the raw parameter θ. However, it can fail when the distribution is periodic or otherwise folded in θ. One possible reason is that a standard neural network conditioning on raw θ has a bias toward simple low-frequency variation in the input coordinate. With limited data, it may treat far-separated but distributionally similar parameters, such as θ and $\theta + T$, as unrelated.

Ultimately, we want likelihood estimation at θ to utilize information from nearby values of θ. Here "nearby" should be interpreted not only in the raw parameter space, but also in the distributional sense: two parameters should be nearby if their conditional distributions have small Hellinger distance. Periodicity is the clearest example: θ is as informative about $p(\cdot \mid \theta)$ as $\theta + T$, but a network fed only raw θ may not exploit this equivalence.

If we already have a rough Hellinger distance estimate, for instance from a binned or kernel-weighted classifier, we can embed the anchor parameters into coordinates $\tilde{\theta}$ in which Euclidean distance better tracks distributional distance. A conditional flow trained on $(\tilde{\theta}, x)$ can then share information between parameters that are close in the estimated distributional geometry. This motivates HeIM-Flow: an iterative procedure that alternates between estimating Hellinger distances with a conditional flow and re-embedding the parameter space with metric multidimensional scaling (MDS).

In practice, HeIM-Flow recovers a Hellinger-distance matrix (and associated decoding quality) that tracks an approximate ground truth well when the observation dimension d is moderate or small, but performance degrades sharply when d is large: estimated matrices and off-diagonal

Pearson correlations to the reference remain strong in 10D yet are substantially weaker in 100D for comparable sample sizes.

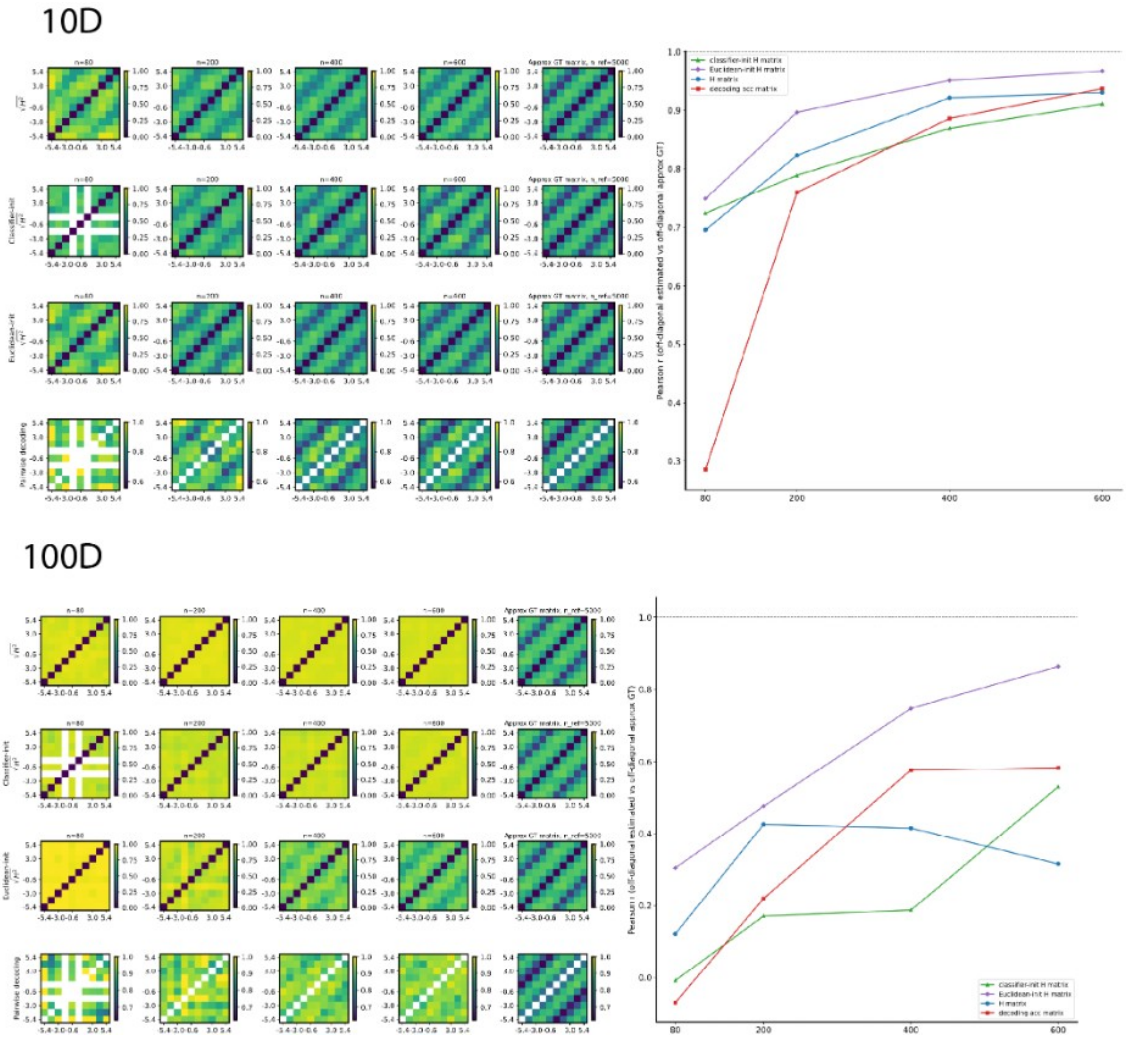

Figure 2: HeIM-Flow in low versus high data dimension, on the same style of binned-$\sqrt{H^2}$ and classifier-based estimates. Top: $d = 10$; bottom: $d = 100$. Left: heatmap grids of estimated distance matrices and pairwise decoding, across sample sizes n, compared to an approximate ground-truth matrix. Right: Pearson correlation of off-diagonal entries between each estimate and the approximate ground truth. In 10D, correlations grow toward near-perfect agreement as n increases; in 100D, they remain much lower, illustrating that the method is effective in low dimension but far less so in a high-dimensional regime.

2.2 Method 1: Estimating log likelihood ratio with flow matching

Let $t \in [0, 1]$ interpolate between a tractable base density p_0 on $\mathbb{R}^d$ (e.g. standard Gaussian) and the model distribution p_1 over observations. The learned time-dependent and stimulus-dependent velocity field $v_t : \mathbb{R}^d \to \mathbb{R}^d$ defines the probability-flow ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}\, x_t(\theta) = v_t(x_t; \theta), \tag{3}$$

whose time-t marginal density we denote p_t. Along a trajectory $\{x_t\}_{t\in[0,1]}$, the instantaneous change-of-variables formula gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\log p_t(x_t \mid \theta) = -\operatorname{div} v_t(x_t;\theta) = -\operatorname{tr}\big(\nabla_x v_t(x_t;\theta)\big), \tag{4}$$

so, writing $x_1 = x$ for a sample from p_1 and x_0 for its preimage under the flow,

$$\log p_1(x \mid \theta) = \log p_0(x_0) - \int_0^1 \operatorname{div} v_t(x_t;\theta)\,\mathrm{d}t. \tag{5}$$

To evaluate $\log p_1(x \mid \theta)$, one integrates the ODE, typically backward from x at $t = 1$ to $x_0(\theta)$ at $t = 0$, and accumulates the divergence along the trajectory.

Once we have $\log p_1(x \mid \theta)$, we can estimate the log-likelihood ratio by

$$\widehat{\ell}_{ij}(x) = \log \widehat{p}(x \mid \theta_i) - \log \widehat{p}(x \mid \theta_j), \tag{6}$$

and then plug $\widehat{\ell}_{ij}(x)$ into (2) to estimate the Hellinger distance.

2.3 Method 2: HeIM-Flow procedure

Let

$$\mathcal{D} = \{(\theta_n, x_n)\}_{n=1}^N, \qquad \theta_n \in \mathbb{R}^m, \qquad x_n \in \mathbb{R}^d. \tag{7}$$

Choose a set of anchor parameters

$$\mathcal{A} = \{\vartheta_i\}_{i=1}^M. \tag{8}$$

The anchors may be all unique stimulus values, a regular grid in parameter space, or a representative subset of the observed θ values. HeIM-Flow estimates a sequence of Hellinger distance matrices

$$D^{(k)} \in \mathbb{R}^{M\times M}, \qquad D_{ij}^{(k)} \approx H(p(x \mid \vartheta_i), p(x \mid \vartheta_j)), \tag{9}$$

and uses each matrix to construct an embedded coordinate system for the next round of flow fitting.

2.3.1 Initialization of the Hellinger distance matrix

We initialize the Hellinger distance matrix using a simple shared-Gaussian approximation. We first divide the parameter space into bins

$$\mathcal{B}_1, \ldots, \mathcal{B}_M,$$

and let

$$I_i = \{n : \theta_n \in \mathcal{B}_i\}$$

denote the samples assigned to bin i. Within each bin, we estimate the conditional mean by

$$\widehat{\mu}_i = \frac{1}{|I_i|}\sum_{n\in I_i} x_n. \tag{10}$$

We assume that the conditional distribution in each bin is approximately Gaussian with bin-dependent mean but shared covariance:

$$p(x \mid \theta \in \mathcal{B}_i) \approx \mathcal{N}(\widehat{\mu}_i, \Sigma). \tag{11}$$

To estimate the shared noise covariance, we compute residuals within each bin,

$$r_n = x_n - \widehat{\mu}_i, \qquad n \in I_i. \tag{12}$$

We then concatenate residuals across all bins and compute the empirical residual covariance

$$S_{\text{res}} = \frac{1}{N-M} \sum_{i=1}^{M} \sum_{n \in I_i} r_n r_n^\top. \tag{13}$$

To obtain a stable precision matrix, we use a Ledoit–Wolf shrinkage estimate of the shared covariance:

$$\widehat{\Sigma}_{\text{LW}} = (1-\rho) S_{\text{res}} + \rho \, \frac{\text{tr}(S_{\text{res}})}{d} I, \tag{14}$$

where the shrinkage coefficient $\rho \in [0,1]$ is chosen analytically by the Ledoit–Wolf procedure. We define the corresponding precision matrix as

$$\widehat{\Lambda}_{\text{LW}} = \widehat{\Sigma}_{\text{LW}}^{-1}. \tag{15}$$

Under the shared-Gaussian approximation, the squared Hellinger distance between bins i and j has the closed form

$$\widehat{H}_{ij}^{2,(0)} = 1 - \exp\left[-\frac{1}{8} (\widehat{\mu}_i - \widehat{\mu}_j)^\top \widehat{\Lambda}_{\text{LW}} (\widehat{\mu}_i - \widehat{\mu}_j) \right]. \tag{16}$$

Therefore, the initial MDS dissimilarity matrix is

$$D_{ij}^{(0)} = \widehat{H}_{ij}^{(0)} = \sqrt{1 - \exp\left[-\frac{1}{8} (\widehat{\mu}_i - \widehat{\mu}_j)^\top \widehat{\Lambda}_{\text{LW}} (\widehat{\mu}_i - \widehat{\mu}_j) \right]}, \qquad D_{ii}^{(0)} = 0. \tag{17}$$

2.3.2 Metric MDS embedding of the anchor parameters

Given the current Hellinger distance matrix $D^{(k)}$, metric MDS finds anchor embeddings

$$z_i^{(k)} \in \mathbb{R}^r, \qquad i = 1, \ldots, M, \tag{18}$$

such that Euclidean distances between the embedded anchors approximate the estimated Hellinger distances. Specifically,

$$Z^{(k)} = \arg \min_{z_1, \ldots, z_M \in \mathbb{R}^r} \sum_{i<j} \left(\|z_i - z_j\|_2 - D_{ij}^{(k)} \right)^2, \tag{19}$$

where $Z^{(k)} \in \mathbb{R}^{M \times r}$ has row vectors $(z_i^{(k)})^\top$. The embedding dimension r is a user-chosen hyperparameter. If θ is one-dimensional but the distribution is periodic, $r = 2$ is often a natural choice because the distributional geometry may be closer to a circle than a line.

2.3.3 Linear interpolation from anchors to training samples

MDS provides embedded coordinates only for the anchors. To train a conditional flow on all samples, we extend the anchor embedding to arbitrary θ by linear interpolation. Denote this interpolation map by

$$g^{(k)} : \mathbb{R}^m \to \mathbb{R}^r, \qquad \tilde{\theta}^{(k)} = g^{(k)}(\theta). \tag{20}$$

For one-dimensional ordered anchors, if

$$\vartheta_i \le \theta \le \vartheta_{i+1}, \tag{21}$$

then

$$g^{(k)}(\theta) = (1-\lambda) z_i^{(k)} + \lambda z_{i+1}^{(k)}, \qquad \lambda = \frac{\theta - \vartheta_i}{\vartheta_{i+1} - \vartheta_i}. \tag{22}$$

If θ is periodic, interpolation is performed circularly so that the last and first anchors are also adjacent. For multi-dimensional grid anchors, the same idea can be implemented with multilinear interpolation. For scattered anchors, one can use simplex-based barycentric interpolation.

The embedded coordinate of each training sample is then

$$\tilde{\theta}_n^{(k)} = g^{(k)}(\theta_n). \tag{23}$$

2.3.4 Flow-based update of the Hellinger distance matrix

At iteration k, train a conditional flow-matching model on the embedded dataset

$$\widetilde{\mathcal{D}}^{(k)} = \{(\tilde{\theta}_n^{(k)}, x_n)\}_{n=1}^N. \tag{24}$$

The specific flow-matching path is not fixed by HeIM-Flow; any path can be used as long as the resulting conditional flow allows evaluation of (see 2.2)

$$\log \widehat{p}_k(x \mid \tilde{\theta}). \tag{25}$$

For anchor pair (i, j), the flow-based log-likelihood ratio is

$$\widehat{\ell}_{ij}^{(k+1)}(x) = \log \widehat{p}_k(x \mid z_i^{(k)}) - \log \widehat{p}_k(x \mid z_j^{(k)}). \tag{26}$$

Once the log-likelihood ratio is available, the Hellinger update follows the sample-wise construction in the Hellinger note. Define

$$\psi(u) = 1 - \operatorname{sech}\left(\frac{u}{2}\right). \tag{27}$$

Because sech is even, $\psi(\widehat{\ell}_{ij}^{(k+1)}(x))$ is the same sample-wise contribution whether the ratio is written as $\log \widehat{p}_i - \log \widehat{p}_j$ or with the opposite sign.

Let I_i denote the samples assigned to the local neighborhood or bin of anchor ϑ_i. The two Monte Carlo averages

$$\widehat{m}_{i\to j}^{(k+1)} = \frac{1}{|I_i|} \sum_{n \in I_i} \psi\left(\widehat{\ell}_{ij}^{(k+1)}(x_n)\right), \qquad \widehat{m}_{j\to i}^{(k+1)} = \frac{1}{|I_j|} \sum_{n \in I_j} \psi\left(\widehat{\ell}_{ij}^{(k+1)}(x_n)\right) \tag{28}$$

estimate the two expectations in the symmetric identity

$$H^2(p_i, p_j) = \frac{1}{2}\mathbb{E}_{X_i \sim p_i}\left[\psi(\ell_{ij}(X_i))\right] + \frac{1}{2}\mathbb{E}_{X_j \sim p_j}\left[\psi(\ell_{ij}(X_j))\right]. \tag{29}$$

Therefore the flow-based squared Hellinger update is

$$\widehat{H}_{ij}^{2,(k+1)} = \frac{1}{2}\widehat{m}_{i\to j}^{(k+1)} + \frac{1}{2}\widehat{m}_{j\to i}^{(k+1)}. \tag{30}$$

After computing all pairs, clip numerical errors and set the diagonal to zero:

$$\widehat{H}_{ij}^{2,(k+1)} \leftarrow \operatorname{clip}\left(\widehat{H}_{ij}^{2,(k+1)}, 0, 1\right), \qquad \widehat{H}_{ii}^{2,(k+1)} = 0. \tag{31}$$

The next MDS dissimilarity matrix is obtained directly from the updated Hellinger estimate:

$$D_{ij}^{(k+1)} = \sqrt{\widehat{H}_{ij}^{2,(k+1)}}. \tag{32}$$

No damping is applied; each iteration uses the newly estimated Hellinger distance matrix directly.

2.3.5 Iterative algorithm

The full HeIM-Flow procedure is:

1. Choose anchor parameters $\{\vartheta_i\}_{i=1}^M$.
2. Assign samples to anchor neighborhoods or bins $I_1, \ldots, I_M$.
3. Initialize $D^{(0)}$ with the shared-Gaussian Hellinger estimate.
4. For $k = 0, 1, \ldots, K-1$:
   (a) Apply metric MDS to $D^{(k)}$ to obtain anchor embeddings $Z^{(k)} = \{z_i^{(k)}\}_{i=1}^M$.
   (b) Construct a linear interpolation map $g^{(k)}$ from ϑ_i to $z_i^{(k)}$.
   (c) Embed all training parameters by $\tilde{\theta}_n^{(k)} = g^{(k)}(\theta_n)$.
   (d) Train a conditional flow-matching model on $\{(\tilde{\theta}_n^{(k)}, x_n)\}_{n=1}^N$.
   (e) Evaluate flow-based log-likelihood ratios using (26) for every anchor pair.
   (f) Convert the log-likelihood ratios into sample-wise contributions with $1 - \operatorname{sech}(\widehat{\ell}/2)$.
   (g) Average those contributions over the two anchor neighborhoods using (30), then set $D^{(k+1)} = \sqrt{\widehat{H}^{2,(k+1)}}$.

The algorithm can be run for a fixed number of iterations or until the Hellinger matrix stabilizes, for example when

$$\frac{\|D^{(k+1)} - D^{(k)}\|_F}{\|D^{(k)}\|_F} \tag{33}$$

becomes sufficiently small.

The output of HeIM-Flow is the final Hellinger distance matrix $D^{(K)}$, the final distributional embedding $Z^{(K)}$, the interpolation map $g^{(K)}$, and the final conditional flow model $\widehat{p}_K(x \mid \tilde{\theta})$.

2.4 Method 3: Estimating Hellinger distance using binary classifiers

We use a classifier-based binned estimator to estimate the Hellinger distance. The procedure has three steps: binning the parameter samples, fitting a balanced binary classifier for each bin pair, and converting the fitted classifier logits into an estimated Hellinger distance.

We first partition the θ domain into fixed equal-width bins and assign each sample to a hard bin index. For each bin pair (i, j), we form a binary classification problem using only samples whose θ values fall into bins i or j. The classifier is trained on the train split only, while the final Hellinger estimate is evaluated on the full subset pool, namely the union of train and validation samples restricted to bins i and j. Validation samples are never used along.

For each bin pair (i, j), we fit a linear logistic classifier

$$f_{ij}(x) = \beta_{ij}^\top x + b_{ij}, \tag{34}$$

where samples from bin i are assigned label 1 and samples from bin j are assigned label 0. Let $\sigma(u) = (1 + e^{-u})^{-1}$. If there are n_i and n_j train samples from bins i and j, respectively, and

$$n_{\text{pair}} = n_i + n_j, \tag{35}$$

then we balance the two classes using sample weights

$$w_i = \frac{1}{2}\frac{n_{\text{pair}}}{n_i}, \qquad w_j = \frac{1}{2}\frac{n_{\text{pair}}}{n_j}. \tag{36}$$

The weighted logistic objective is therefore

$$\mathcal{L}_{ij}^{\text{wlogit}}(\beta_{ij}, b_{ij}) = -\sum_{x\in\mathcal{X}_i^{\text{tr}}} w_i \log\sigma(f_{ij}(x)) - \sum_{x\in\mathcal{X}_j^{\text{tr}}} w_j \log(1-\sigma(f_{ij}(x)))\,, \tag{37}$$

where $\mathcal{X}_i^{\text{tr}}$ and $\mathcal{X}_j^{\text{tr}}$ denote the train-split samples from bins i and j. This weighting makes the two classes contribute equally to the classifier fit, because

$$n_i w_i = n_j w_j = \frac{1}{2} n_{\text{pair}}. \tag{38}$$

At the same time,

$$n_i w_i + n_j w_j = n_{\text{pair}}, \tag{39}$$

After fitting the classifier, we use its logit as an estimate of the log-likelihood ratio between the two bin-conditional distributions:

$$\widehat{\ell}_{ij}(x) = \text{logit}\ \widehat{c}_{ij}(x) = f_{ij}(x). \tag{40}$$

Using the symmetric Hellinger transform $\psi(u) = 1 - \text{sech}(u/2)$, we estimate the squared Hellinger distance between bins i and j by evaluating the fitted classifier on the full-pool samples from the two bins:

$$\widehat{H}^2_{ij,\text{clf}} = \frac{1}{2}\frac{1}{|\mathcal{X}_i|}\sum_{x\in\mathcal{X}_i}\psi\Big(\widehat{\ell}_{ij}(x)\Big) + \frac{1}{2}\frac{1}{|\mathcal{X}_j|}\sum_{x\in\mathcal{X}_j}\psi\Big(\widehat{\ell}_{ij}(x)\Big)\,, \tag{41}$$

where $\mathcal{X}_i$ and $\mathcal{X}_j$ denote the full-pool samples in bins i and j. The class-balancing weights are used only during classifier fitting. They affect the fitted parameters β_{ij} and b_{ij}, and hence the estimated logit $\widehat{\ell}_{ij}(x)$, but the Hellinger estimate itself is computed as an unweighted within-bin average over the evaluation pool.

Finally, we symmetrize and clip the estimated squared Hellinger matrix:

$$\widehat{H}^2_{ij,\text{clf}} \leftarrow \text{clip}\left(\frac{\widehat{H}^2_{ij,\text{clf}} + \widehat{H}^2_{ji,\text{clf}}}{2}, 0, 1\right), \qquad \widehat{H}^2_{ii,\text{clf}} = 0. \tag{42}$$

For visualization and comparison, we use the corresponding Hellinger distance

$$\widehat{D}^{\text{clf}}_{ij} = \sqrt{\widehat{H}^2_{ij,\text{clf}}}. \tag{43}$$